

Analysis group meeting

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Cooking for Pass0v9

Cooking for mini-timelines, Pass0v9_dcv2, started Sunday 23 February.

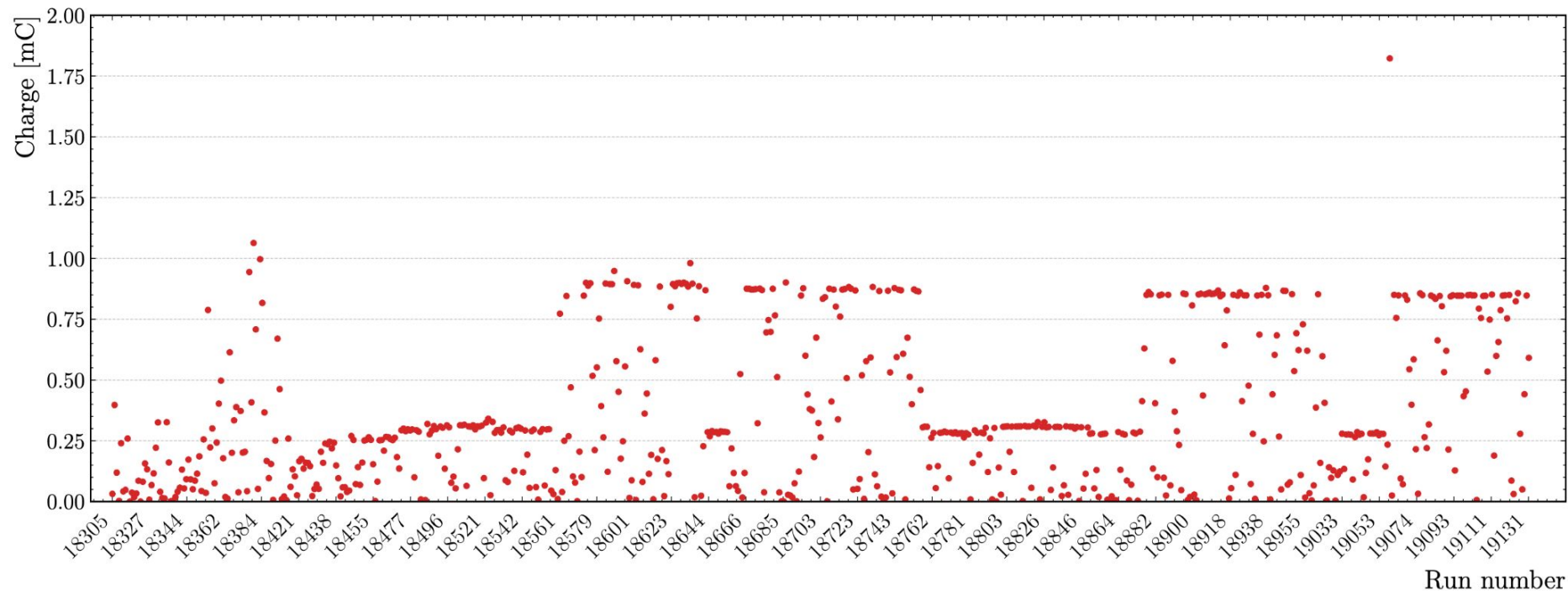
Name	Complete	Jobs	
rgd-rqa-pass0v9_dcv2_run_018316-18316	100.00%	15	
rgd-rqa-pass0v9_dcv2_CxC-18339x14	100.00%	210	
rgd-rqa-pass0v9_dcv2_SER_LD2-18327x4	100.00%	60	
rgd-rqa-pass0v9_dcv2_SER_s-18482x5	100.00%	75	
rgd-rqa-pass0v9_dcv2_LD2-18318x17	98.82%	255	
rgd-rqa-pass0v9_dcv2_CuSn-18347x22	95.15%	330	

I have skimmed the pass0v9;

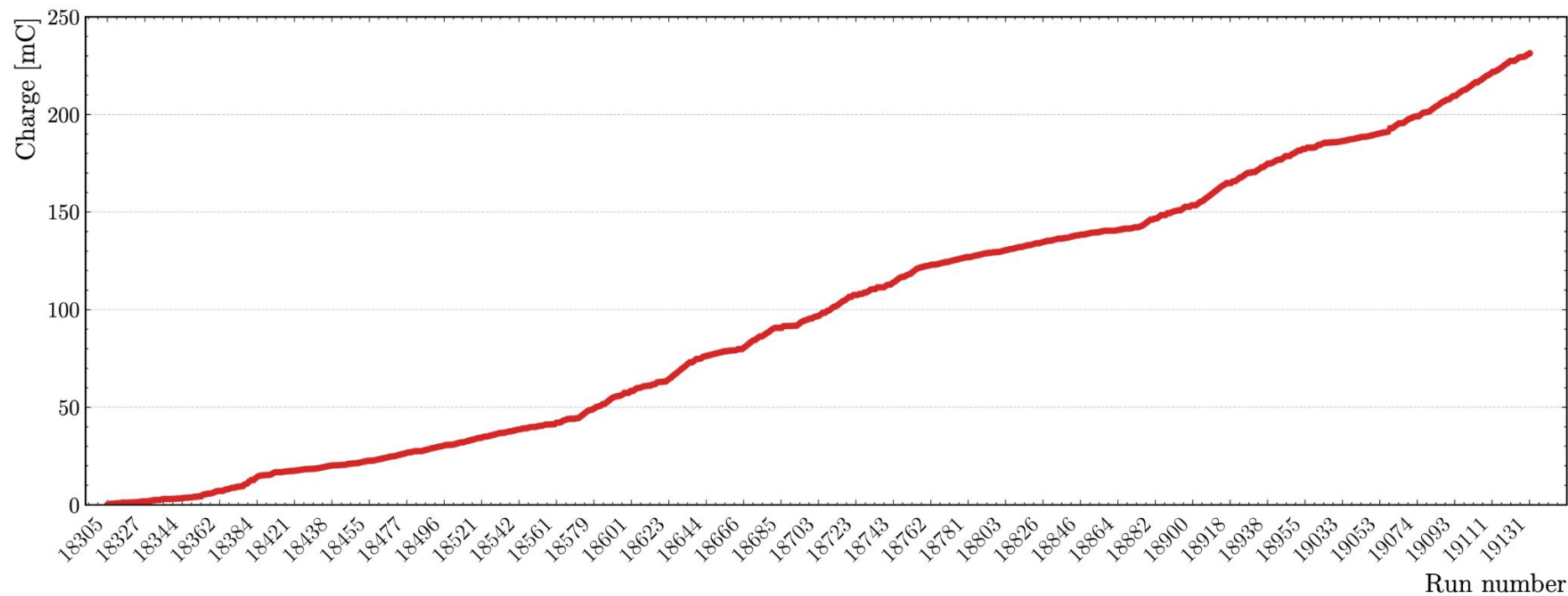
The files are available on /cache/hallb/scratch directory

Check timelines, faraday cup charge

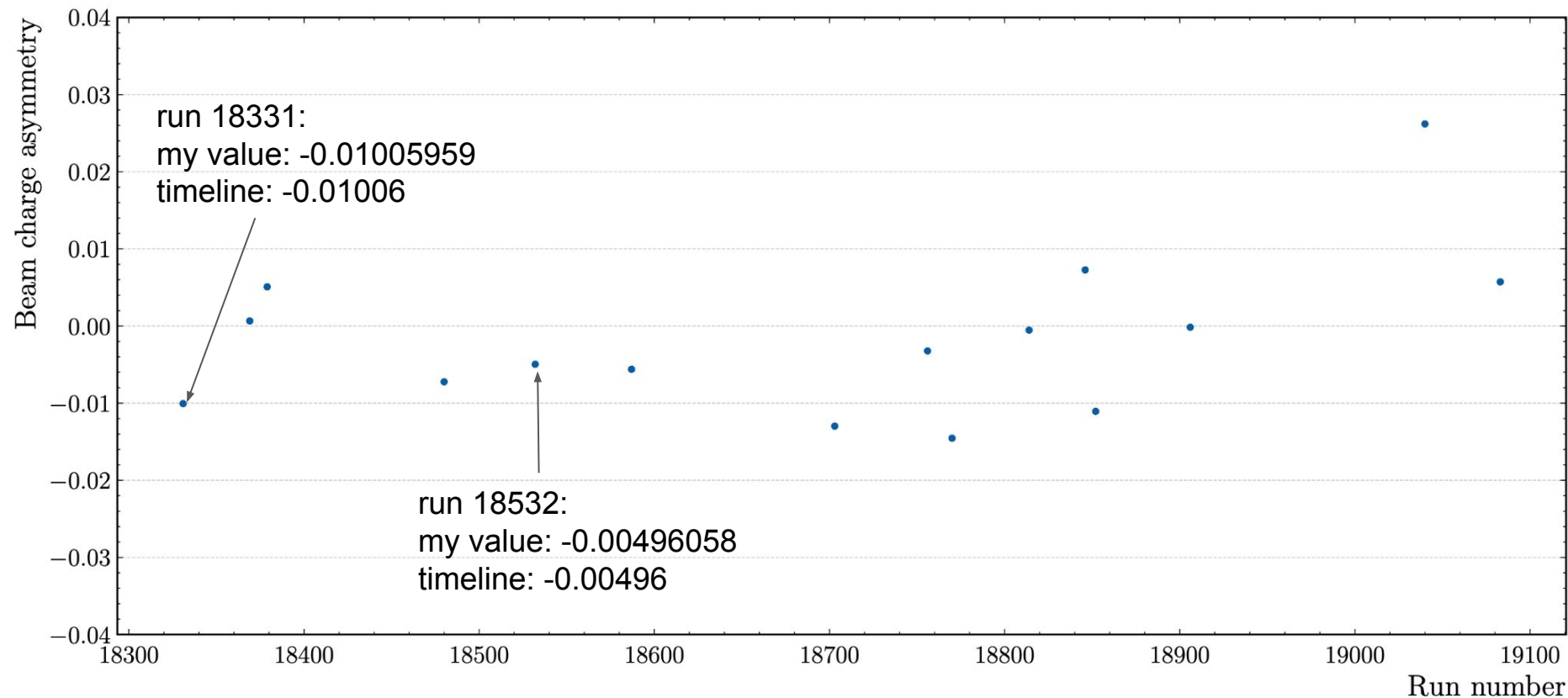
3



Check timelines, faraday cup charge accumulated



Check timelines, beam charge asymmetry



Run Number	Charge	Sector	nTrigger	My Value	Timelines
18532	0.28352866	1	188953	0.666434	0.66643
		2	186635	0.658258	0.65826
		3	188220	0.663848	0.66385
		4	180457	0.636468	0.063647
		5	184638	0.651215	0.65121
		6	177411	0.625725	0.62573

Check timelines, electron FD normalized yield

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Run Number	Charge [nC]	Sector	nTrigger	My Value [1/nC]	Timelines [1/nC]
18331	0.325550362	1	48,458	0.148849	0.14885
		2	42,969	0.131989	0.13199
		3	39,778	0.122187	0.12219
		4	47,976	0.147369	0.14737
		5	41,075	0.126171	0.12737
		6	37,752	0.115964	0.11596

First, the runs is split in 3 bins, first bin is the first event, second bin is between the first event to the last RUN::Scaler bank with tag set at 1, and third one is from the last bank to the end.

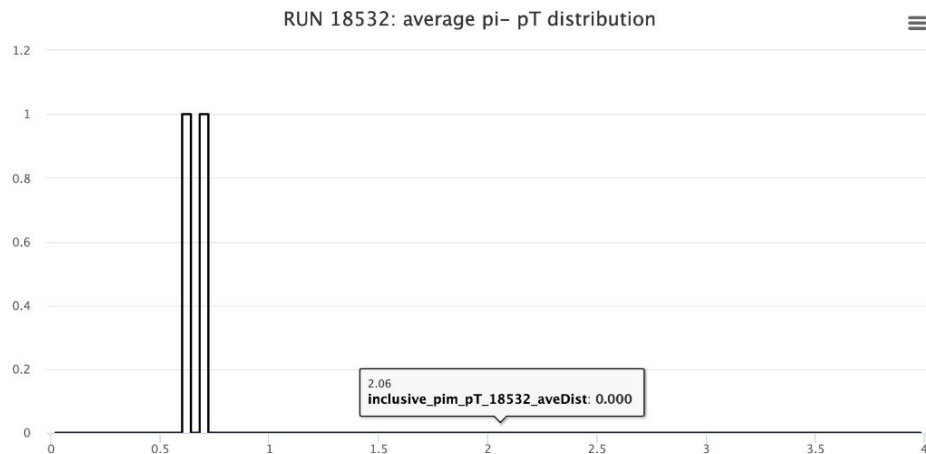
We need to fill a histogram with 50 bins to get the mean.

After, we need to fill another histogram with 100 bins with the means of each bins.

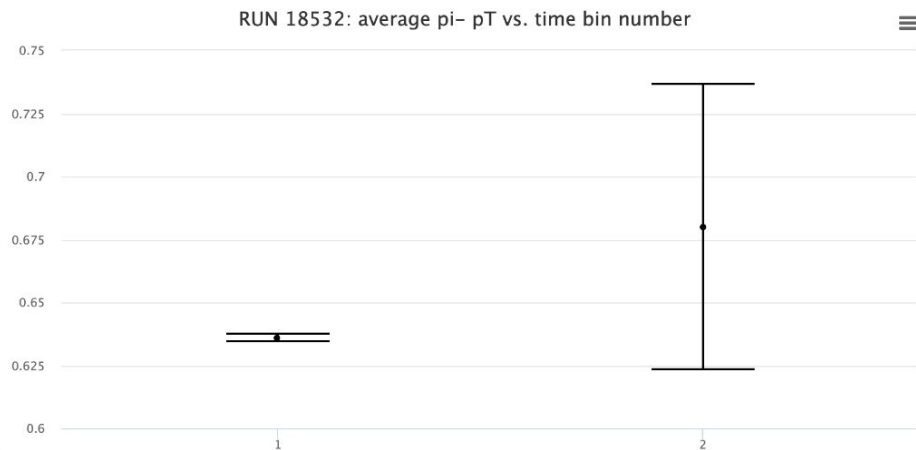
After that we need to compute the mean of these means.

Check timelines, piMinus kinematics

The histogram with 100 bins fill
with the means

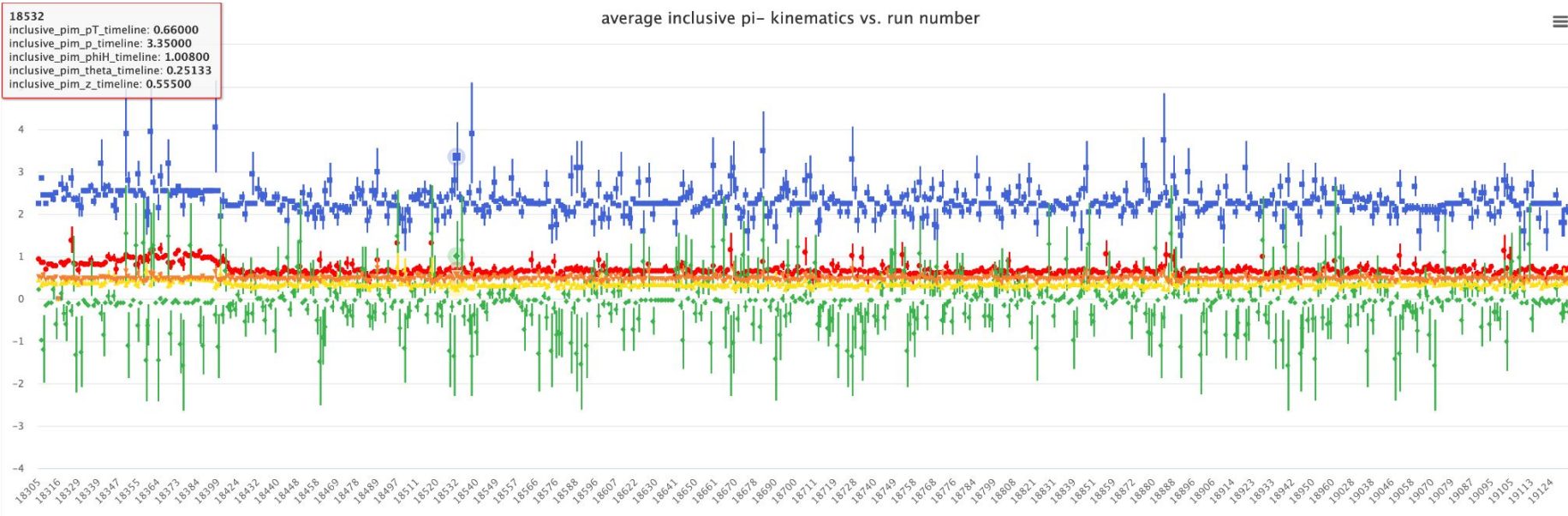


The mean of the histogram with
50 bins



Check timelines, piMinus kinematics

The mean of all the means



Check timelines, piMinus kinematics

Run 18532: Bin 1 is always empty

Bin 2 mine | timelines

kinematics p mean: 2.25 | 2.25 GeV

kinematics pT mean: 0.62 | 0.62 GeV

kinematics z mean: 0.47500000000000003 | 0.475

kinematics theta mean: 0.32201317999999995 | 0.32201317999999 rad

kinematics phih mean: -0.094499999999999989 | -0.09449999999999998 rad

Bin 3

kinematics p mean: 4.45 | 4.45 GeV

kinematics pT mean: 0.70000000000000001 | 0.7 GeV

kinematics z mean: 0.635 | 0.635

kinematics theta mean: 0.180641540000000002 | 0.18064154 rad

kinematics phih mean: 2.1105 | 2.1105 rad

Check timelines, piMinus kinematics

Mean: $p = 3.35 \mid 3.35 \text{ GeV}$

$p_T = 0.66 \mid 0.66 \text{ GeV}$

$\phi_H = 1.008 \mid 1.008 \text{ rad}$

$\theta = 0.25132736 \mid 0.25133 \text{ rad}$

$z = 0.555 \mid 0.555$

Work for this run, but I have some rounding error:

For example I have a momentum of 1.900000000001 GeV, it goes in the bin 1.95 GeV but the timelines say it should be 1.85 GeV. It is due to the fact it is in reality 0.19.

So for RG-D the first bin is empty, the second bin have almost all the events and the third bins have 1 or 2 events.

So taking the mean of each bins is bad for us since we take the mean for only two bins and one of them is bad (for example ϕ_H).

It is not an issue for other run group because they have more bins, so one bad is not a huge issue

- Work on the presentations, both the AI for ALERT and RG-D update.
- RG-D analysis:
 - Work on the timelines to fix the rounding error and compute beam spin asymmetry
 - Cross check for the nuclear transparency vs Q^2 , find why the nuclear transparency for Cu and Sn is so high.
 - Work on π^0 analysis: Fix the event mixing method
- Background merging, generate 100k events for one configuration
- Chris ask me to copy the high-QA timelines to the RG-D directory, is it ok?
- Chaden Djalali ask me to swap shifts, is it ok with you?

Date	Expert Owl (00:00-08:00)	Expert Day (08:00-16:00)	Expert Swing (16:00-24:00)	Worker Day (07:00-15:00)	Worker Swing (15:00-23:00)	Worker Owl (23:00-07:00)	Accelerator Schedule/ Hall B Program
27-Apr-2025 Sunday	Timothy Hayward	Stepan Stepanyan	Lamiaa El Fassi	Buddhiman Tamang	Chaden Djalali	Kenichi Nakano	RG-L
28-Apr-2025 Monday	Timothy Hayward	Stepan Stepanyan	Lamiaa El Fassi	Buddhiman Tamang	Chaden Djalali	Kenichi Nakano	RG-L
29-Apr-2025 Tuesday	FIU	JMU	Yordanka Ilieva	Buddhiman Tamang	Chaden Djalali	Kenichi Nakano	RG-L
30-Apr-2025 Wednesday	FIU	JMU	Yordanka Ilieva	Buddhiman Tamang	Chaden Djalali	Kenichi Nakano	RG-L
28-Jul-2025 Monday	CNU	Eric Voutier	Igor Strakovsky	Asar Ahmed	Mathieu Ouillon	Gregory Matousek	RG-L
29-Jul-2025 Tuesday	CNU	Eric Voutier	Shift	Chao Peng	Mathieu Ouillon	Gregory Matousek	RG-L
30-Jul-2025 Wednesday	Valerio Mascagna	Eric Fuchey	GSIFFN	Chao Peng	Mathieu Ouillon	Gregory Matousek	RG-L
31-Jul-2025 Thursday	Valerio Mascagna	Eric Fuchey	GSIFFN	Chao Peng	Mathieu Ouillon	Gregory Matousek	RG-L

Handwritten annotations: A double-headed arrow connects the first two rows of the top table. Another double-headed arrow connects the first two rows of the bottom table, with the word "swap" and "??" written next to it.